INTERNATIONAL JOURNAL OF INFORMATION SECURITY SCIENCE
M. Priyadharshini, Vol.15, No.3, pp.1-17
https://doi.org/10.55859/ijiss.1795996

A Secure Logic Locking Integration Technique for AES Algorithm

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Research Paper Received: 03.10.2025 Revised: 25.05.2026 Accepted: 31.07.2026

Abstract—The globalization of semiconductor manufacturing has increased vulnerabilities such as hardware Trojan insertion, overbuilding, reverse engineering, and intellectual property piracy. Logic locking has emerged as an effective countermeasure to protect integrated circuits using secret key activation. The Advanced Encryption Standard (AES) is widely deployed in Internet of Things (IoT) systems and therefore requires robust hardware-level protection. This work presents an architecture-level integration of Random Logic Locking and an Anti-SAT block within an AES encryption core implemented on an Artix-7 FPGA. Unlike prior studies that primarily evaluate Anti-SAT on benchmark circuits, the proposed approach embeds the locking framework directly into a complete cryptographic datapath and analyzes both hardware overhead and security characteristics. The design is evaluated in terms of area, power, throughput, Hamming distance corruptibility, and Satisfiability Checking(SAT) attack resilience. Experimental results demonstrate approximately 50% output corruption under incorrect keys and incorrect key convergence during SAT attack attempts across multiple key sizes. The results indicate that the proposed integration enhances SAT attack resistance while maintaining acceptable hardware overhead, making it suitable for secure IoT hardware deployments.

Keywords—logic locking, SAT, Hardware Trojans, Random Locking, Anti-SAT

1. Introduction

The ever-increasing cost and complexity of maintaining a foundry with advanced technologies have made the semiconductor industries ‘fabless.’ The processes involved in the Integrated Circuit (IC) supply chain happen worldwide, so the IC is prone to many kinds of hardware attacks like insertion of Hardware Trojans (HT), counterfeiting ICs, reverse engineering, IC overbuilding, Intellectual Property (IP) piracy, and side-channel attacks. An adversary or attacker can claim ownership by reverse engineering an IP/IC or overbuilding ICs and making a profit illegally. The insertion of malicious components, the HTs are implanted to acquire secret information, password theft, denial of service, etc. These kinds of attacks pave the way to billions of losses to the semiconductor industries. Various countermeasures against these attacks have been explored: IC camouflaging, split manufacturing, IC metering, IP watermarking, physical unclonable functions, etc.

Logic Locking (LL) is a versatile countermeasure that protects the design from attacks at all phases

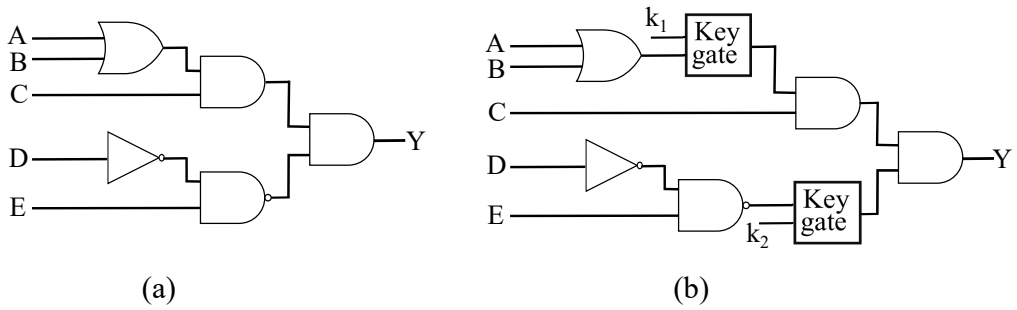


Figure 1. Example of logic locking (a) original design (b) locked design with key gate insertions

in the IC design cycle [1]. LL is a technique to lock the original design with a secret key for which additional key gates are inserted. The key gates lock the circuit and only when the correct key input is fed the design produces the right functionality. Ending Piracy of Integrated Circuits (EPIC) [2] is a foremost locking technique that protects the chips at the foundry by uniquely locking each IC with a key. In addition, EPIC brings in a combinational Random Logic Locking (RLL) method using public key cryptography. To ensure better corruptibility at the output for a wrong key, fault analysis-based logic encryption techniques [3], [4] were developed. Logic encryption is related to fault propagation and ensures that the wrong key's effect for different inputs is always propagated to the output. The techniques also aim to target for 50% hamming distance between the correct and incorrect outputs on the application of a wrong key. This fault analysis-based logic encryption is achieved with minimal gate additions.

The attack strategies from an attacker's point of view were analyzed and a logic obfuscation technique [5] based on key-gate interference was developed. A built-in self-authentication (BISA) technique [6], has been developed to eliminate the spare spaces in the circuit layout by filling it with functional fillers, leaving no space for trojan-gate insertions. This circuitry is also applied to different benchmark designs. A logic encryption technique [7] to prevent overproduction and avoid insertion of HTs by minimizing the number of signals with low controllability. The HTs are usually inserted in the low controllability nodes, hence the attackers or adversary will not be able to exploit the real rare values in a circuit. The technique [8] performs encryption that increases the triggering probability of rare nets and also reduces the number of vulnerable nets with minimum gate replacements.

Various novel locking schemes are proposed and as a challenge to each technique, attack methods are explored and these works are reviewed [1], [9], [10], [11], [12] in detail by considering their merits and demerits. Many researchers have studied the shortcomings of the existing combinational LL techniques and proposed some algorithmic attack scenarios like sensitization [5], SAT [13], and circuit partitioning attacks [14] to retrieve the key information. Also, approximate attacks like AppSAT [15] and Double-DIP [16], structural attacks like signal probability skew attack [17], AppSAT guided removal attack [18], Bypass attack [19], the Structural Attack on Logic Locking (SAIL) [20], and side-channel attacks [21], [22] have been explored in the past. The SAT attack was a breakthrough that could break the secret key within a few hours for many of the combinational LL techniques [2], [4], [5], [7], [23].

The rest of the article is organized as follows: In Section 2 we present the discussion and works related to logic locking, SAT attack, and a brief about the AES algorithm. Section 3 explains the security analysis of the RLL technique and in detail the proposed integration technique. Section 4 discusses the experimental results and its analysis based on design overhead and SAT attack resistance, section 5 extends the proposed technique to the AES algorithm, and finally section 6 concludes the work with references.

2. Preliminaries and Related Work

2.1. Logic Locking

Logic Locking is a technique that inserts additional key gates and key inputs to safeguard the functionality of a design. LL helps to protect the IC design from hardware attacks. Various LL techniques have been explored in the past. They are classified based on the different types of key gates and the location of key-gate insertion algorithms. According to the type of key gates, they are classified as XOR/XNOR key-gate [2], [4], [5], MUX-based [4], [14], [24], [25], and Look-up Table(LUT)-based [23], [26], [27], [28] insertions. Also, AND/OR key gate insertions [7] and logic cones [29] have also been studied. There are different types of insertion algorithms like RLL [2], where key gates are inserted randomly in the design, and fault analysis-based logic locking [4], [23] where key gates are inserted based on fault-propagation and maximum corruptibility at primary outputs. The secure logic locking [5] aimed for maximum interference between key gates such that the correct key cannot be rightly sensitized to the primary outputs. An example combinational circuit and a logically locked circuit are depicted in Fig. 1 (a) and (b) respectively. The circuit is logically locked with the secret keys k_1 and k_2. Only on the correct application of the key inputs the circuit produces the correct output. The main objective of these traditional LL methods was to increase output corruptibility when an incorrect key was fed. Also, obfuscation techniques for digital signal processing applications using high-level transformation like folding [30] and obfuscation for a trustworthy scan chain for wireless networks [31] have been developed in the past.

2.2. SAT Attack

A SAT algorithm-based attack model was proposed by Pramod et al [13] to analyze and explore the security of existing combinational LL algorithms. SAT attack was a generalized attack that was able to retrieve the secret key for many benchmark circuits with different logic locking techniques. SAT attack efficiently obtained the correct key for 95% of the encrypted circuits and for 90% of the circuits required only 250 or fewer input/output observations. The attack model carefully finds input patterns that iteratively rule out equivalence keys. The special inputs and outputs obtained from the functional IC are referred to as Distinguishing Input/Output (DIO) pairs. The SAT algorithm also finds the correct secret key which is consistent with all the DIO patterns.

By analyzing the strengths and weaknesses of the SAT attack, researchers have proposed many techniques, namely SAT-Attack Resistant Logic Locking (SARLock)[32], Anti-SAT [33], Tenacious and Traceless logic Locking (TTLock) [34], Stripped-Functionality Logic Locking - Hamming Distance (SFLL-HD) [35], cyclic obfuscation [36], Dynamically-Obfuscated Scan Chain(DOSC) [37], lightweight Anti-SAT [38], Strong Anti-SAT (SAS) [39],Generalized (G-Anti-SAT) [40] to overcome the existing attacks. The SARLock tries to itera-

tively maximize the DIO patterns required to thwart the correct secret key. SARLock aims to exponentially improve the effort involved in tracing the number of secret key bits, but overhead increases linearly. The Anti-SAT block, a relatively lightweight circuit block was designed to mitigate the SAT attack. A set of keys was attached to the original design to lock the circuit, and another set was linked to the Anti-SAT block. This helps to increase the total number of iterations exponentially and the time required to find the correct secret key when the Anti-SAT block is integrated into the original design. Lockit [41] is an attempt to automate the logic locking flow based on the SFLL-HD technique. Probability-based Logic-Locking (ProbLock) [42] is a probability-based LL technique for both combinational and sequential circuits, wherein the technique was also evaluated against the SAT attack and proved to consume a longer duration to acquire the secret key. The work [43] is a Secure Interference Logic Locking (SILL) method that provides SAT attack resistance and less area overhead for combinational circuits.

Replacement-Based Key-Controlled Circuits(RKC) [44] is a lightweight logic locking approach in which selected gates are replaced with key-controlled structures to lock the circuit. This technique achieves SAT attack resilience against classical SAT attack and its variants at the cost of low area and performance overhead. The work [45] is a depth-coupled logic locking(DCLL) technique where the secret key is embedded deep across multiple logic levels to increase the SAT solver complexity with the design overhead on a slightly higher side. GateLock [46] is a SAT attack-resistant logic locking technique where input patterns are partitioned, and multiple key sequences dependent on inputs are used to lock the benchmark circuits. The enhanced framework is a work [47] which embeds an Anti-SAT block with decoy keys and a challenge-response authentication system to increase the attacker's time in the SAT-based key extraction process. Even though the extracted key seems correct, it only works for a limited number of challenges. Recent studies such as

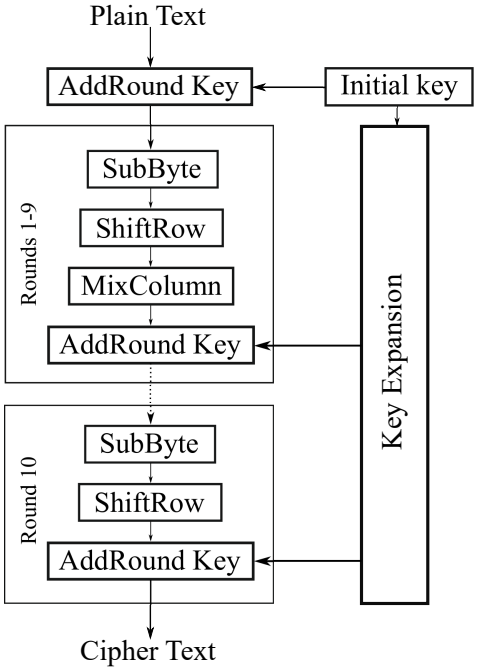


Figure 2. Outline of AES encryption architecture

Locking Enabled Differential Analysis (LEDA) [48], Locking Enabled Differential Fault Analysis (LEDFA) [48], and divide-and-conquer SAT-based cryptanalytic approaches [49] have highlighted potential vulnerabilities associated with applying conventional RLL to cryptographic circuits. In particular, uniformly distributed XOR/XNOR key-gate insertion can perturb critical AES transformations and potentially expose structurally sensitive regions to differential, fault-assisted, or partition-based analyses. These studies emphasize that weak key-gate interaction and indiscriminate key placement may adversely affect the security characteristics of cryptographic implementations.

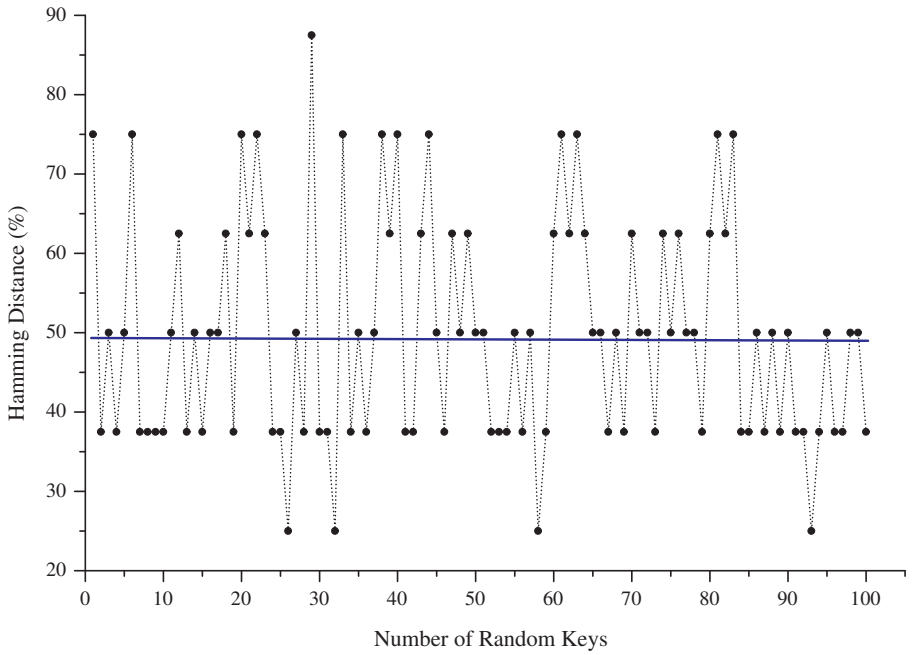


Figure 3. Hamming distance between correct and incorrect outputs

To mitigate these limitations, the proposed work employs module-aware key placement integrated with Anti-SAT entanglement across AES functional modules instead of purely uniform insertion. This increases key dependency and output corruptibility while reducing localized structural exposure and limiting effective key convergence under SAT-based attacks. The major contributions of this proposed work are as follows:

1 To integrate RLL with Anti-SAT block specifically for the AES architecture, instead of applying Anti-SAT or logic locking in isolation.
2 To design and evaluate a full AES datapath, rather than on generic benchmarks, improving applicability for real-time cryptographic IP protection.
3 To implement and validate experimentally the proposed design on the Artix-7 Field Programmable Gate Array(FPGA), which provides hardware-level evidence and feasibility in an acceptable range of area, power overhead.
4 The security validation is conducted by the hamming distance corruptibility which achieves a 50% output corruption for incorrect keys.
5 The SAT attack resiliency is demonstrated for the locked AES Substitution-box design, where the retrieved keys were found incorrect for varying key lengths.

2.3. AES-Advanced Encryption Standard

The Rijndael algorithm was selected as the Advanced Encryption Standard (AES) by the National Institute of Standards and Technology (NIST) in 2001 to replace the old Data Encryption Standard (DES). The AES is an iterative, symmetric block cipher algorithm with the same key for encryption

and decryption. The size of the data block is fixed with a cipher key of 128, 192, or 256-bit length. This proposed work considers a key bit size of 128 bits. The input and output of the AES algorithm both constitute a sequence of 128 bits in binary value. The architecture of the AES algorithm for encryption is illustrated in Fig. 2.

The AES algorithm for encryption consists of four byte-oriented transformations namely the SubBytes, ShiftRows, MixColumns, and AddRoundKey in every round except the final round which does not have the MixColumns. There is an initial round key addition followed by implementing the round function ten times. The cipher key goes through the key expansion routine to generate a key for each round. The Substitution box (S-box) is the main core block of the AES algorithm as it is a non-linear structure substituting each byte of the state with a value from the S-box. Hence the proposed technique targets to implement a secure S-box circuit design for the AES algorithm. In the past, different types of obfuscation techniques [50], [51], [52] for the encryption of AES-128 have been explored.

Table 1.
Hamming distance and area overhead for RLL of S-box design

Keysize	Area overhead (%)	Hamming distance (%)
3	1.2	46.42
4	1.6	46.67
5	2	45.56
6	2.4	45.35
7	2.8 (≈3%)	49.62

3. Proposed Methodology

Once a design is logically locked, any wrong key entered should produce a wrong output. Any output depends on both the secret key and input test pattern. Also, how many bits are corrupted because of the wrong key is the main point of concern with respect to logic locking. Hamming distance is a key metric that denotes the mismatch between any two binary strings of equal length. Any secret key and input plain text can either produce an output with a hamming distance between correct and incorrect output as zero when none of the bits are corrupted or can produce a hamming distance of 100 with all the output bits corrupted on the application of a wrong key. Both these scenarios help the attacker to extract the key quite easily since the output will be either in its true or complemented form. Therefore, a hamming distance of around 50% is considered ideal since the secret key would be difficult to trace from the corrupted output bits.

3.1. Security Analysis of RLL in S-box design

RLL is a technique where the key gates are inserted in a random manner throughout the entire design. RLL provides better coverage of gates all over the design. In this proposed method initially, the Composite Field Arithmetic(CFA) [53] based S-box design is subjected to RLL. A combination of XOR/XNOR gates with inverters are used to lock the circuit. The number of bits getting corrupted at the output due to an incorrect secret key varies based on the key size of the secret key. Fig. 3 shows the hamming distance between the correct and incorrect key values when 100 random key values are applied. An average hamming distance of 49.62% (≈50%) is achieved by performing the conventional RLL technique.

Table 1 shows the area overhead (in terms of gate count) and hamming distance when RLL is performed for the S-box design by varying the key bit size. The Hamming distance approaches 50% when the key size for random locking is varied from

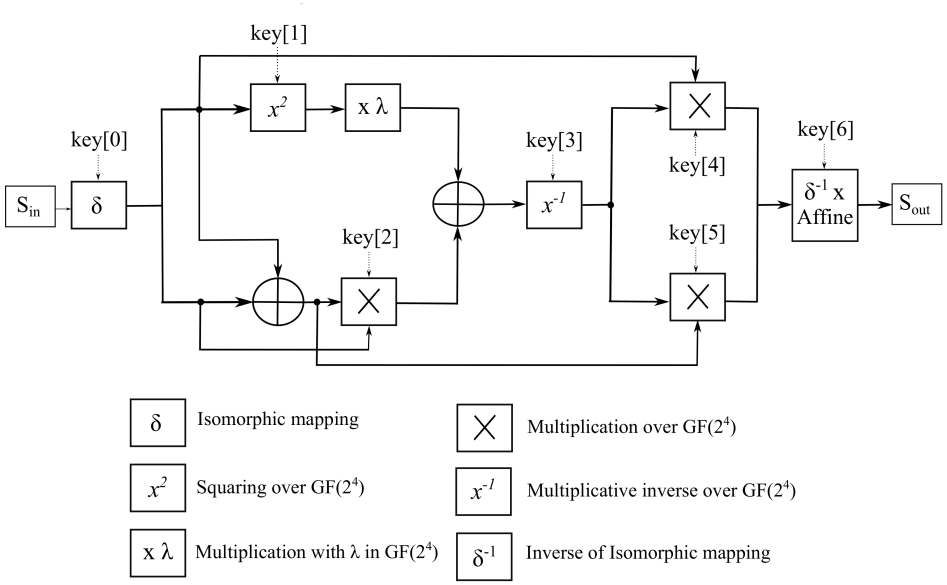


Figure 4. Randomly locked architecture of S-box

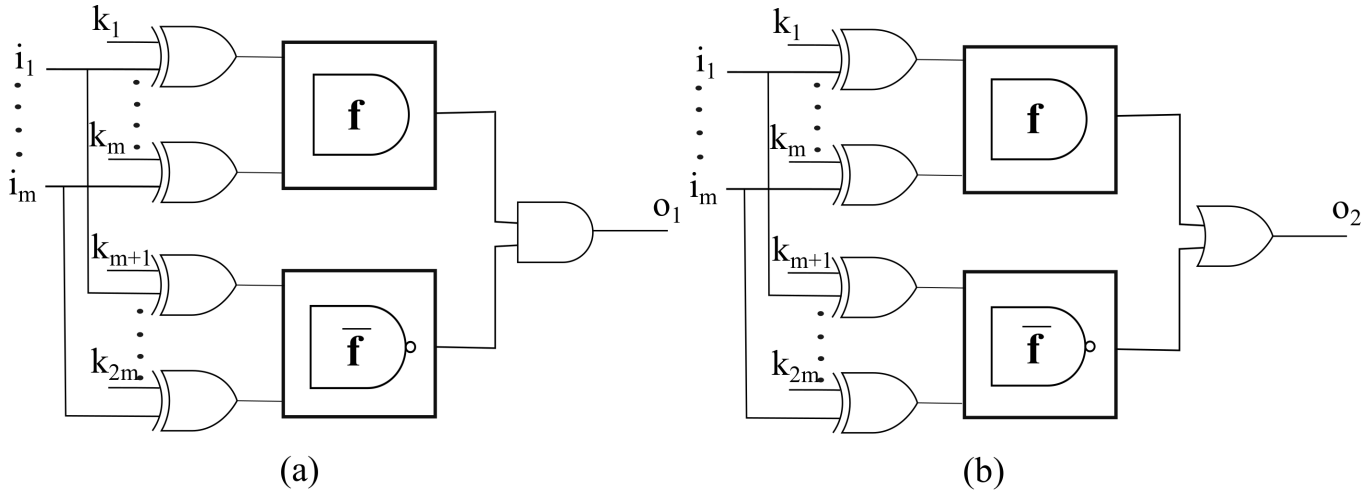


Figure 5. Anti-SAT block configuration (a) Type-0 (b) Type-1

3 to 7. The ideal hamming distance for logic locking is achieved and the area overhead is ≈3% which is a very nominal value for RLL. Considering these criteria, the key bit size of 7 is further not increased and fixed for random locking. Fig .4 shows the architecture of the randomly locked S-box design where the XOR/XNOR key gates are connected internally in the different subblocks. The S-box has an 8-bit input (S_{in}) and 8-bit output (S_{out}) with a 7-bit random locking key (key[0:7]).

3.2. Anti-SAT Block Design

3.2.1 Need for Anti-SAT design

In this methodology, RLL is performed and the Anti-SAT block is integrated into the S-box block. The S-box design operates independently on each byte of the state. The Anti-SAT block

is a lightweight block that can be integrated into any original circuit design and is mathematically proven resilient against the SAT attack. Hence by including the Anti-SAT block in the S-box design the time duration to go through for each byte of data extends the number of iterations. This integration will produce a SAT resistant logically locked design. This serves as a motivation behind choosing the Anti-SAT block to integrate with RLL to achieve a secure locking technique. Although, RLL may exhibit limited key-gate interference when applied with purely uniform placement, it offers low hardware overhead and implementation simplicity. In this work, RLL is employed as a lightweight locking layer, while the Anti-SAT block provides strong SAT resilience, forming a hybrid locking strategy.

3.2.2 Anti-SAT Block

In the traditional logic locking techniques, just increasing the size of the key will not improve the resistance against the SAT attack. The Anti-SAT method exhibits its efficiency in nullifying the SAT attack as computationally infeasible. Anti-SAT aims to increase the number of iterations thereby the solving time required to trace the key. The Anti-SAT design basically has two logic blocks f and $\bar{f}$ that are complements to each other. The two configurations of the anti-SAT block type-0 and type-1 are depicted in Fig.5(a) and 5(b) respectively. The inputs to the two logic blocks are from a set of XOR gates that share the same set of inputs and keys $(k_1, k_2, ...k_m)$ and $(k_{m+1}, ...k_{2m})$ respectively. In this proposed work, f and $\bar{f}$ are chosen as m-input AND and NAND gates respectively, since they result in an exponential complexity of SAT attack with m [33]. The output of f and $\bar{f}$ are fed to an AND gate for type-0 Anti-SAT and an OR gate for type -1 Anti-SAT. Since f and $\bar{f}$ are complements when a correct key is fed the output O_1 will be a zero and output O_2 will be 1 always. This is subject to change based on incorrect key values and input value combinations.

3.2.3 Integrating Anti-SAT with RLL

The conventional logic locking techniques target to cause functionality deviation whenever an incorrect key is fed. The Anti-SAT block aims to increase the SAT attack complexity but does not target functionality deviation. These are the two main reasons to perform the integration of an Anti-SAT block with RLL for a critical application component of AES like S-box. The outline of the proposed integration technique is shown in Fig.6. The CFA architecture of the S-box design is subjected to RLL where an n-bit key is used to lock the circuit. This block is integrated with the Anti-SAT block using an XOR gate for type-0 and an XNOR gate for type-1 configuration. The key size of the Anti-SAT block is $2m$, thus the integrated architecture is locked with a logic locking key (LLK) of size $n+2m$. The S-box is an 8-bit input (S_{in}) and 8-bit output (S_{out}) circuit design where all the primary inputs are fed as input to the XOR gates of the Anti-SAT block. XNOR gates or a combination of XOR/XNOR gates with inverters can replace the XOR gates in the Anti-SAT block. So these inverters will be sometimes bubble pushed to other gates based on the logic functionality and thus the attacker cannot easily extract the key gates.

A detailed comparative analysis with different LL techniques [41] and constraints [54] shows that RLL provides resilience against removal attacks and Anti-SAT provides resilience against sensitization and SAT attacks. Also, RLL ensures high corruptibility at the output, making integrating RLL and Anti-SAT an optimal choice for this proposed work. The area and power overhead also balance to an

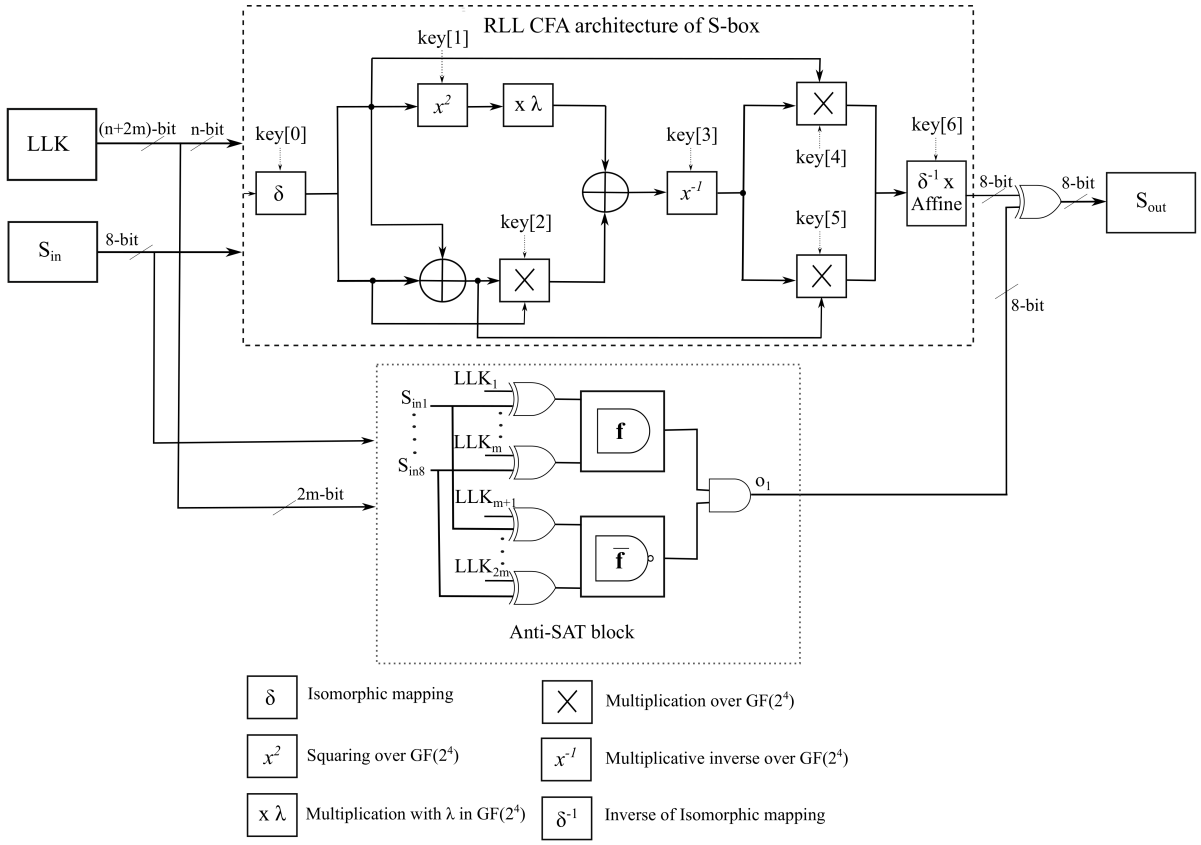


Figure 6. Proposed Integrated Architecture for the S-box Design

Table 2.
FPGA Resource Utilization for Logically Locked S-box Design

Resources	**Available**	**Category**	**Estimated**	**Utilization** %	**Resource Overhead %**
LUT	20800	S-box	32	0.15	-
		S-box+RLL	54	0.26	0.11
		S-box+RLL+Anti-SAT	70	0.34	0.19
IO	106	S-box	16	15.09	-
		S-box+RLL	25	23.58	8.49
		S-box+RLL+Anti-SAT	59	55.56	40.47

acceptable range for this integration. In order to have a securely locked circuit we combine the 3% RLL + Anti-SAT. The proposed Anti-SAT integrated logic locked mechanism aims for a design that substantially elevates the computational complexity of classical SAT-based attacks, by making the key extraction impractical in reasonal time bounds.

The proposed design distributes key gates across multiple AES functional modules instead of purely uniform insertion. This creates functional depen-

dencies between key bits and cryptographic transformations, thereby reducing structural inference and improving resistance to inference-based attacks such as SAIL [20]. Approximate SAT attacks such as AppSAT can partially recover XOR/XNOR key bits in certain designs. However, the cryptographic nature of AES requires exact functionality, and the integrated Anti-SAT block introduces complementary logic that produces significant output corruption under incorrect keys, preventing functional approximation and preserving security [15].

4. Experimental Results and Performance Analysis

The proposed integration technique is described in Verilog hardware descriptive language and implemented on the Artix-7 FPGA. The Xilinx Vivado 2019.1 software is used to simulate, synthesize, and implement the design. The computer system used for this work has 64GB of RAM with a 64-bit operating system.

The experimental results are analyzed in the below perspectives:

1 Design overhead
2 SAT attack resiliency

4.1. Design overhead

The synthesized results of the proposed technique for the logically locked S-box design are detailed in this section. The extra combinational logic used to perform the random locking and integrate the anti-SAT module will add to the overall design. The resource utilization details are illustrated in Table 2. This shows the utilization, the percentage of utilization of available resources, and the overhead in each case. It is found that the overhead is 0.11% and 0.19% for RLL and RLL+Anti-SAT respectively. The throughput is also analyzed in this experiment. The throughput is calculated as the ratio of maximum clock frequency (F_{max}) to latency (the number of clock cycles L) multiplied by 8, which will be the data transferred per clock cycle. The proposed technique achieves a high throughput value of 0.68 Gbps with minimum design overhead for the anti-SAT integrated S-box design.

$$T = (\frac{F_{max}}{L}) \times 8 \tag{1}$$

Table 3.
ASIC implementation results for S-box design

Parameters	**S-box**	**S-box+RLL**	**S-box+RLL +Anti-SAT**
Area (GE)	345	418	505
Power (mW)	0.42	0.63	0.64

The area and power values of the original and the proposed integrated designs in Application-Specific Integrated Circuit(ASIC) platform are estimated using the CADENCE tool, gpdk90nm technology library. Table 3 presents the area and power values. It is observed that the S-box design when logically locked using the combinational logic contributes to the additional area and power overhead. The area overhead for the RLL+Anti-SAT integrated technique is 46.3%. The power overhead for the RLL and RLL+Anti-SAT techniques are 50% and 52.4% respectively.

4.2. SAT attack resiliency

SAT attack was able to break the secret key of many of the existing combinational logic locking techniques. In this proposed technique, to have better security, output corruptibility, and resistance against SAT attack, the S-box design is subjected to the integration of RLL + anti-SAT.

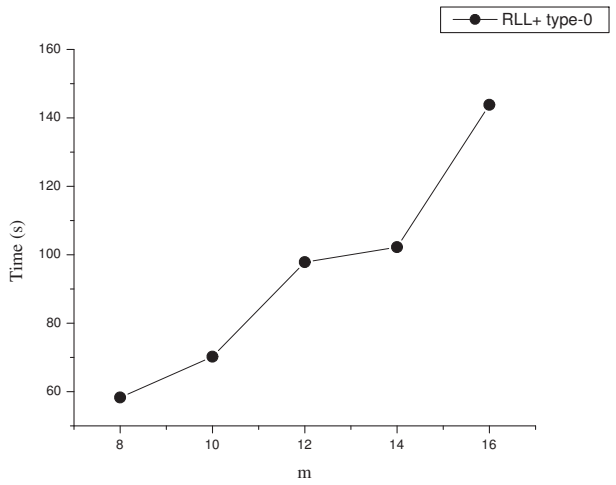


Figure 7. Variation of time with size of Anti-SAT block

4.2.1 Case 1: Correct key extracted

A gate overhead of 3% is considered for performing the random locking and it is observed that the 7-bit secret key is extracted by the SAT attack.

- Expected key : 1010_101
- Found key : 1010_101

The expected and the extracted keys match exactly in three iterations. For an 8-bit S-box design, the secret key size is increased starting from 2-bit up to a maximum of 7-bit key. The SAT attack is capable of tracking the key in minimum iterations.

4.2.2 Case 2: Incorrect key extracted

The anti-SAT block is integrated into the locked S-box circuit by varying the key bit length. The XOR gates included in the anti-SAT block are a combination of XOR+NOT or XNOR+NOT gates. The key bit may vary in size accordingly, where r represents the area overhead in RLL and m represents the number of inputs to the f and $\bar{f}$ blocks of the anti-SAT design. In Table 4 the number of iterations, time duration, and hamming distance between the key values extracted are tabulated for both type-o and type-1 configurations of the Anti-SAT block. It is observed that the number of iterations for RLL+type-0 is 256 iterations in all the cases, and the time increases linearly as seen in Fig.7. The hamming distance mentioned is the variation between the expected correct key value of the integrated design and the extracted key by the SAT attack. It is observed that none of the key values matches and that is evident from the hamming distance values. Even though the time is in a tolerable range, the secret key produced by the SAT attack technique is wrong in all the cases for both type-0 and type-1 configurations. It is to be noted that the number of iterations for type-1, m=8 is only 3. The number of iterations always depends on the SAT solver based on how the DIO pairs are chosen. Also, the time varies based on the host computer system configuration. The following is the expected correct key value and the secret key extracted by the SAT attack.

- Expected key:1010_1010_1000_0010_1001_ 0010_0111_1111_0010_0111_110
- Found key :1001_0011_1111_1000_0001_ 1110_1001_1001_0110_1001_010

This is the key for m=8 type-0 configuration. The key length varies because of the extra NOT gates included. It is found that the percentage of the hamming distance between the key value varies in all the cases of different m values. Thus the proposed integration technique provides better security and SAT attack resistance. In this proposed technique, for the S-box design the type-0 configuration seems to produce more number of iterations and time duration. However, this depends on the original design and SAT solver based on how the DIO pairs are chosen.

Table 4.
Impact of the size of anti-SAT block on SAT attack

Configuration	m	8	10	12	14	16
RLL+Type-0	Iterations	256	256	256	256	256
	Time (s)	58.29	70.22	97.84	102.21	143.81
	Hamming Distance (%)	51.1	45.1	35.6	50.7	45.3
RLL+Type-1	Iterations	3	256	256	256	256
	Time (s)	8.20	57.80	86.02	99.10	132.21
	Hamming Distance (%)	37.2	45.1	40	38.3	64

Table 5.
FPGA Resource Utilization Summary of Logically Locked AES Algorithm

Resources	Utilization (%)			Resource Overhead (%)
	Original AES	AES+RLL	AES+RLL+ Anti-SAT	
LUT	16.52	24.09	30	13.48
FF	5.05	5.05	5.05	0

Table 6.
Hamming distance analysis for random locked AES design

Keysize	Hamming distance (%)
9	47.7
16	50.6
24	55.2
32	60.8

For any Anti-SAT block, with c primary inputs the total number of input combinations is 2^c, so the maximum number of iterations to find the DIO pairs will be less than or equal to 2^c. Hence we can see the maximum number of iterations for this 8-bit AES S-box is 256. However, by integrating the Anti-SAT block into the original design the design is resistant to SAT attack. The secret key extracted in each case for different m values is an incorrect key which provides a secure design.

5. Results for the AES Encryption Algorithm

The proposed integration technique is further extended to the entire AES algorithm by performing the random locking technique to the overall AES design with the anti-SAT engulfed S-box block. The anti-SAT engulfed S-box design when involved in the 9 rounds of the encryption algorithm adds to the overall design overhead. The resource utilization for the RLL and anti-SAT block is shown in Table 5. The additional overhead in the logically locked design is a compromise for the security against the SAT attack. The key size for RLL of the S-box design was fixed to 7-bit which resulted in an expected hamming distance of 50%. The hamming

Table 7.
Comparison with existing obfuscation techniques for AES design

Parameters	[51]	[29]	[50]	Proposed
Area Overhead	4.22%	3.43%	0.88%	13.48%
Throughput	4.35 Gbps	2.34 Gbps	4.25 Gbps	4.34 Gbps
SAT attack resilient	No	No	No	Yes
Hamming Distance	50%	-	50%	50%
Operation	Key-based Obfuscation	Netlist-based Obfuscation	Key-based Obfuscation	Anti-SAT +RLL
Platform	Artix-7	Virtex-7	Artix-7	Artix-7

distance by performing RLL for the AES design is evaluated for different sizes of secret keys. This is tabulated in Table 6. It is observed that the hamming distance increases with size and a key size of 16-bit approaches a hamming distance close to 50%. The RLL ensures the security of the AES algorithm by locking the different transformations involved in the design. This proposed method delivers a high-resilience logic locking framework against SAT-based attacks for the AES architecture.

6. Comparative Analysis

Table 7 presents a comparison of the proposed technique with existing obfuscation-based approaches reported in the literature for FPGA-based AES implementations. The comparison focuses on AES-specific security mechanisms, implementation characteristics, and security-related performance metrics. The proposed technique is advantageous with a comparable throughput value and resilience against the SAT attack. However area overhead will have to be compromised for this attack-resistant integration technique. The obfuscation techniques consider the whole design as a black box whereas in logic locking there will be the interference of keys inside each transformation and each round of the AES design. Hence this contributes to the overall increase in the overhead.

Table 8 summarises representative logic locking techniques evaluated through ASIC-based implementations. The comparison primarily focuses on implementation overhead and security-related metrics reported for ASIC-based logic locking approaches. SARLock is also a technique developed to combat the SAT attack. However, in SARLock increasing the hamming distance to ideal still remains a future scope for study. The traditional RLL and Strong Logic Locking(SLL) techniques are vulnerable to the SAT attack. The best use of these traditional techniques is made by integrating them with the Anti-SAT block. Anti-SAT can be configured and scaled by varying the m values based on the original design of the study. This is the major advantage of the Anti-SAT block, unlike the other techniques. The circuits for study however vary in each technique, the proposed technique seems to produce less area and power overhead than the other logic locking techniques. Also, the hamming distance of 50% is achieved from a security perspective.

Considering the SAT-attack-resistant techniques [44], [45], [46], [47] the GateLock technique relies on input dependency, and RKC relies on selective gate replacement. However, the proposed

Table 8.
Comparison with existing ASIC-Based Logic Locking Techniques

Parameters	RLL[2]	SLL[5]	SARLock+SLL[32]	Proposed RLL +Anti-SAT
Area overhead (%)	29.6	32.2	35.2	**25.0**
Power overhead (%)	45.0	59.2	61.0	**28.7**
Hamming Distance (%)	-	-	-	**50**
SAT attack resilient	No	No	Yes	**Yes**

Table 9.
Comparison with existing SAT-resistant techniques

Technique	Core Idea	SAT resistance	Hardware Overhead	Design Complexity	Key Strength
GateLock [46]	Input-dependent locked gates with multiple logical keys	High (classical SAT)	Very Low	Moderate	Low overhead, input-driven protection
Enhanced Anti-SAT [47]	Anti-SAT + decoy keys + challenge–response	Very High	Moderate	High	Strong SAT hardness
DCLL [45]	Deep multi-level key embedding	High	Moderate –High	High	Structural SAT complexity increase
RKC [44]	Gate replacement using key-controlled logic	Moderate –High	Low	Low–Moderate	Lightweight implementation
Proposed Technique	Integrated RLL + Anti-SAT tailored to AES	High (Experimentally verified) SAT failure)	Low –Moderate (FPGA validated)	Moderate	Balanced security–overhead trade-off with AES-specific implementation

work achieves stronger SAT resistance by explicitly integrating an Anti-SAT block. Even though the enhanced Anti-SAT and DCLL offer high SAT resistance, they occupy higher design complexity and overhead. Table 9 presents a comparative analysis of existing SAT-resistant logic locking techniques. The comparison highlights the underlying locking mechanisms, SAT resistance strategies, and associated design overheads with respect to the proposed approach. The proposed technique offers a balanced trade-off between security and overhead without deep structural entanglement or heavy decoy logic for AES-specific implementation.

The AES S-box is known to be vulnerable to side-channel and fault-assisted attacks such as Simple Power Analysis (SPA), Differential Power Analysis (DPA), and Differential Fault Analysis (DFA). The primary objective of this work is to enhance protection against reverse engineering and classical SAT-based attacks using logic locking. Dedicated countermeasures against side-channel and fault-assisted

attacks, such as masking, hiding, or fault detection mechanisms, are not incorporated in the current implementation and remain as future work. However, the additional locking logic may introduce switching activity variations that can potentially affect side-channel observations.

7. Conclusion

Logic locking provides an effective hardware security mechanism against reverse engineering, counterfeiting, and IP piracy. This work presented an integrated logic locking framework for the AES architecture implemented on an Artix-7 FPGA by combining Random Logic Locking with an Anti-SAT block. The proposed approach was evaluated using hardware metrics including area, power, and throughput, along with security analysis based on Hamming distance corruptibility and SAT attack behavior. Experimental results showed approximately 50% output corruption under incorrect keys and incorrect key convergence during SAT attack attempts for different key sizes, indicating improved resistance to key extraction. The hardware overhead remains within practical limits for FPGA-based secure applications. Overall, the study demonstrates a practical architecture-level locking strategy that enhances the security of AES implementations while preserving implementation feasibility.

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